

trgn515_assign2_sub_github

ranzhu

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Task 1: Indel Calling - Create a Slurm Job

Illumina

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --mem=16G
#SBATCH --time=6:00:00
#SBATCH --job-name=indel_calling_illumina
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/indel_calling_illumina_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/indel_calling_illumina_%j.err

source ~/.bashrc
export OPENBLAS_NUM_THREADS=1
conda activate /project2/msalomon_1816/trgn_515/conda_env/trgn515

ref_genome=/project2/msalomon_1816/trgn_515/2_sv/ref_genome/GRCh38/genome.fa
input_bam=/project2/msalomon_1816/ranzhu/assignment_1/mapping/illumina_chr20.realigned.bam
out_dir=/project2/msalomon_1816/ranzhu/assignment_2/02_indels/illumina
output=$out_dir/illumina.chr20.indels.filtered.vcf.gz
mkdir -p $out_dir

bcftools mpileup --threads 4 \
    -f $ref_genome \
    -a AD,ADF,ADR,DP,SP \
    -q 0 \
    -Q 0 \
    -Ou \
    $input_bam \
| bcftools call \
    -m \
    -v \
    -V snps \
    -Ou \
| bcftools norm \
    -f $ref_genome \
```

```

-m -any \
-Ou \
| bcftools filter -m+ -s 'MinMQ' -e 'MQ < 20' \
| bcftools filter -m+ -s 'MinQUAL' -e 'QUAL < 20' \
| bcftools filter -m+ -s 'MinDP' -e 'FMT/DP < 20' \
| bcftools filter -m+ -s 'MinAD' \
-e '(FMT/GT ~ "1/1" && FMT/AD[:1] < 10) || (FMT/GT ~ "0/1" && FMT/AD[:1] < 5)' \
-Oz -o $output --threads 4

tabix -p vcf $output

```

Pacbio

```

#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=16
#SBATCH --mem=64G
#SBATCH --time=24:00:00
#SBATCH --job-name=indel_calling_pacbio
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/indel_calling_pacbio_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/indel_calling_pacbio_%j.err

set -euo pipefail

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/clair

export OPENBLAS_NUM_THREADS=1
export OMP_NUM_THREADS=1

which bcftools
which tabix
bcftools --version

model=/project2/msalomon_1816/trgn_515/conda_env/clair/Clair3/models/hifi
ref_genome=/project2/msalomon_1816/trgn_515/2_sv/ref_genome/GRCh38/genome.fa
input_bam=/project2/msalomon_1816/ranzhu/assignment_1/mapping/pacbio.chr20.sorted.bam
out_dir=/project2/msalomon_1816/ranzhu/assignment_2/02_indels/pacbio

mkdir -p $out_dir

run_clair3.sh \
--bam_fn=$input_bam \
--ref_fn=$ref_genome \
--threads=$SLURM_CPUS_PER_TASK \
--platform="hifi" \
--model_path=$model \
--output=$out_dir

```

```

bcftools view \
--types indels \
$out_dir/merge_output.vcf.gz \
-Ou \
| bcftools norm \
--threads $SLURM_CPUS_PER_TASK \
-f $ref_genome \
-Ou \
| bcftools filter -m+ -s 'MinDP' -e 'FMT/DP < 20' \
| bcftools filter -m+ -s 'MinAD' -e '(FMT/GT ~ "1/1" && FMT/AD[0:1] < 10) || (FMT/GT ~ "0/1" && FMT/AD[0:1] < 10)' \
-Oz --threads $SLURM_CPUS_PER_TASK \
-o $out_dir/pacbio.chr20.indels.filtered.vcf.gz

tabix -p vcf $out_dir/pacbio.chr20.indels.filtered.vcf.gz

```

ONT

```

#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=16
#SBATCH --mem=64G
#SBATCH --time=24:00:00
#SBATCH --job-name=indel_calling_ont
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/indel_calling_ont_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/indel_calling_ont_%j.err

source ~/.bashrc
export OPENBLAS_NUM_THREADS=1
conda activate /project2/msalomon_1816/trgn_515/conda_env/clair

which bcftools
which tabix
bcftools --version

model=/project2/msalomon_1816/trgn_515/conda_env/clair/Clair3/models/r941_prom_hac_g360+g422
ref_genome=/project2/msalomon_1816/trgn_515/2_sv/ref_genome/GRCh38/genome.fa
input_bam=/project2/msalomon_1816/ranzhu/assignment_1/mapping/ont.chr20.sorted.bam
out_dir=/project2/msalomon_1816/ranzhu/assignment_2/02_indels/ont

mkdir -p $out_dir

run_clair3.sh \
--bam_fn=$input_bam \
--ref_fn=$ref_genome \
--threads=$SLURM_CPUS_PER_TASK \
--platform="ont" \
--model_path=$model \
--output=$out_dir

```

```

bcftools view \
--types indels \
$out_dir/merge_output.vcf.gz \
-Ou \
| bcftools norm \
--threads $SLURM_CPUS_PER_TASK \
-f $ref_genome \
-Ou \
| bcftools filter -m+ -s 'MinDP' -e 'FMT/DP < 20' -Ou \
| bcftools filter -m+ -s 'MinAD' -e '(FMT/GT ~ "1/1" && FMT/AD[0:1] < 10) || (FMT/GT ~ "0/1" && FMT/AD[0:1] < 10)' -Ou \
-Oz --threads $SLURM_CPUS_PER_TASK \
-o $out_dir/ont.chr20.indels.filtered.vcf.gz

tabix -p vcf $out_dir/ont.chr20.indels.filtered.vcf.gz

```

Task 2: Structural Variant calling

1. Create a slurm job

illumina-mapping with GRCh37

```

#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --job-name=sv_illumina_delly
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/sv_illumina_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/sv_illumina_%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/trgn515

FASTQ1=/project2/msalomon_1816/ranzhu/assignment_1/fastq/hg200_illumina_chr20_1.fastq.gz
FASTQ2=/project2/msalomon_1816/ranzhu/assignment_1/fastq/hg200_illumina_chr20_2.fastq.gz

REF_BWA=/project2/biodb/genomes/Homo_sapiens/NCBI/build37.2/Sequence/BWAIndex/genome.fa
REF_FASTA=/project2/msalomon_1816/trgn_515/2_sv/ref_genome/GRCh37/genome.fa

OUTDIR=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/illumina_chr20
mkdir -p $OUTDIR

SAMPLE=hg200

flowcell=$(zcat "$FASTQ1" | head -n1 | cut -d':' -f3)
lane=$(zcat "$FASTQ1" | head -n1 | cut -d':' -f4)

```

```

RGID=${flowcell}.${lane}
PU=${flowcell}.${lane}.${SAMPLE}
PL=ILLUMINA

bwa mem -t 8 \
  -R "@RG\tID:${RGID}\tPU:${PU}\tPL:${PL}\tSM:${SAMPLE}" \
  "$REF_BWA" \
  "$FASTQ1" \
  "$FASTQ2" \
  | samtools view -bS - \
  | samtools sort -@ 8 -o "$OUTDIR/illumina_chr20.GRCh37.sorted.bam"
samtools index "$OUTDIR/illumina_chr20.GRCh37.sorted.bam"

```

illumina-delly_sv

```

#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --job-name=illumina_delly
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/illumina_delly%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/illumina_delly%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/delly

FASTQ1=/project2/msalomon_1816/ranzhu/assignment_1/fastq/hg200_illumina_chr20_1.fastq.gz
FASTQ2=/project2/msalomon_1816/ranzhu/assignment_1/fastq/hg200_illumina_chr20_2.fastq.gz

REF_BWA=/project2/biodb/genomes/Homo_sapiens/NCBI/build37.2/Sequence/BWAIndex/genome.fa
REF_FASTA=/project2/msalomon_1816/trgn_515/2_sv/ref_genome/GRCh37/genome.fa

OUTDIR=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/illumina_chr20
mkdir -p $OUTDIR

delly call \
-g $REF_FASTA \
-o $OUTDIR/illumina_chr20.sv.bcf \
$OUTDIR/illumina_chr20.GRCh37.sorted.bam

```

illumina-convert and filter

```

#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main

```

```

#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --job-name=sv_illumina_1
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/sv_illumina_1%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/sv_illumina_1%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/trgn515

OUTDIR=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/illumina_chr20
mkdir -p $OUTDIR

bcftools view \
$OUTDIR/illumina_chr20.sv.bcf \
-Oz -o $OUTDIR/illumina_chr20.sv.vcf.gz

bcftools view \
-f PASS \
-i 'INFO/SVTYPE="DEL" || INFO/SVTYPE="INS"' \
$OUTDIR/illumina_chr20.sv.vcf.gz \
-Oz -o $OUTDIR/illumina_chr20.sv.pass.vcf.gz

bcftools index $OUTDIR/illumina_chr20.sv.pass.vcf.gz

```

ONT-mapping with GRCh37

```

#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --job-name=ont_map
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/ont_map_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/ont_map_%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/trgn515

FASTQ=/project2/msalomon_1816/ranzhu/assignment_1/fastq/hg200_ont_chr20.fastq.gz
IN=/project2/msalomon_1816/ranzhu/assignment_1/fastq
OUT=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/ont_chr20GRCh37
REF=/project2/msalomon_1816/trgn_515/2_sv/ref_genome/GRCh37/genome.fa
mkdir -p $OUT

CHR=20

```

```

minimap2 -ax map-ont -t 4 \
  $REF \
  $IN/hg200_ont_chr20.fastq.gz \
  | samtools view -bS - \
  | samtools sort -@ 4 -o $OUT/ont_chr20.GRCh37.sorted.bam

samtools index $OUT/ont_chr20.GRCh37.sorted.bam

```

ONT-sniffles_sv

```

#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --job-name=ont_delly
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/ont_delly%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/ont_delly%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/sniffles

REF=/project2/msalomon_1816/trgn_515/2_sv/ref_genome/GRCh37/genome.fa
BAM=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/ont_chr20GRCh37/ont_chr20.GRCh37.sorted.bam
OUT=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/ont_chr20GRCh37/
mkdir -p $OUT

sniffles \
  --input $BAM \
  --vcf $OUT/ont_chr20.sniffles.vcf.gz \
  --reference $REF \
  --threads 8

```

ONT-filter

```

#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --job-name=ont_fliter
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/ont_fliter_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/ont_fliter_%j.err

```

```

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/trgn515

OUT=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/ont_chr20GRCh37

bcftools view -f PASS \
    $OUT/ont_chr20.sniffles.vcf.gz \
    -Oz -o $OUT/ont_chr20.sniffles.pass.vcf.gz

bcftools index $OUT/ont_chr20.sniffles.pass.vcf.gz

bcftools view \
    -i 'INFO/SVTYPE="DEL" || INFO/SVTYPE="INS"' \
    $OUT/ont_chr20.sniffles.pass.vcf.gz \
    -Oz -o $OUT/ont_chr20.sniffles.pass.DEL_INS.vcf.gz

bcftools index $OUT/ont_chr20.sniffles.pass.DEL_INS.vcf.gz

```

Pacbio-mapping with GRCh37

```

#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --job-name=pacbio_map
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/pacbio_map_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/pacbio_map_%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/trgn515

FASTQ=/project2/msalomon_1816/ranzhu/assignment_1/fastq/hg200_pacbio_chr20.fastq.gz
IN=/project2/msalomon_1816/ranzhu/assignment_1/fastq
OUT=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/pacbio_chr20
REF=/project2/msalomon_1816/trgn_515/2_sv/ref_genome/GRCh37/genome.fa
mkdir -p $OUT

CHR=20

minimap2 -ax map-hifi -t 4 \
    $REF \
    $IN/hg200_pacbio_chr20.fastq.gz \
    | samtools view -bS - \
    | samtools sort -@ 4 -o $OUT/pacbio_chr20.GRCh37.sorted.bam

samtools index $OUT/pacbio_chr20.GRCh37.sorted.bam

```

Pacbio-sniffles_sv

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --job-name=sv_illumina_delly
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/sv_illumina_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/sv_illumina_%j.err

module load bwa
module load samtools
source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/delly

FASTQ1=/project2/msalomon_1816/ranzhu/assignment_1/fastq/hg200_illumina_chr20_1.fastq.gz
FASTQ2=/project2/msalomon_1816/ranzhu/assignment_1/fastq/hg200_illumina_chr20_2.fastq.gz

REF_BWA=/project2/biodb/genomes/Homo_sapiens/NCBI/build37.2/Sequence/BWAIndex/genome.fa
REF_FASTA=/project2/msalomon_1816/trgn_515/2_sv/ref_genome/GRCh37/genome.fa

OUTDIR=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/illumina_chr20
mkdir -p $OUTDIR

SAMPLE=hg200

flowcell=$(zcat "$FASTQ1" | head -n1 | cut -d':' -f3)
lane=$(zcat "$FASTQ1" | head -n1 | cut -d':' -f4)

RGID=${flowcell}.${lane}
PU=${flowcell}.${lane}.${SAMPLE}
PL=ILLUMINA

bwa mem -t 8 \
  -R "@RG\tID:${RGID}\tPU:${PU}\tPL:${PL}\tSM:${SAMPLE}" \
  "$REF_BWA" \
  "$FASTQ1" \
  "$FASTQ2" \
  | samtools view -bS - \
  | samtools sort -@ 8 -o "$OUTDIR/illumina_chr20.GRCh37.sorted.bam"
samtools index "$OUTDIR/illumina_chr20.GRCh37.sorted.bam"

delly call \
  -g $REF_FASTA \
  -o $OUTDIR/illumina_chr20.sv.bcf \
  $OUTDIR/illumina_chr20.GRCh37.sorted.bam

bcftools view \
  $OUTDIR/illumina_chr20.sv.bcf \
```

```

-0z -o $OUTDIR/illumina_chr20.sv.vcf.gz

bcftools view \
-f PASS \
$OUTDIR/illumina_chr20.sv.vcf.gz \
-0z -o $OUTDIR/illumina_chr20.sv.pass.vcf.gz

bcftools index $OUTDIR/illumina_chr20.sv.pass.vcf.gz

```

Pacbio-filter

```

#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --job-name=pacbio_filter
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/pacbio_filter_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/pacbio_filter_%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/trgn515

OUTDIR=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/pacbio_chr20

bcftools view \
-f PASS \
$OUTDIR/pacbio_chr20.sniffles.vcf.gz \
-0z -o $OUTDIR/pacbio_chr20.sniffles.pass.vcf.gz

bcftools index $OUTDIR/pacbio_chr20.sniffles.pass.vcf.gz

bcftools view \
-i 'INFO/SVTYPE="DEL" || INFO/SVTYPE="INS"' \
$OUTDIR/pacbio_chr20.sniffles.pass.vcf.gz \
-0z -o $OUTDIR/pacbio_chr20.sniffles.pass.DEL_INS.vcf.gz

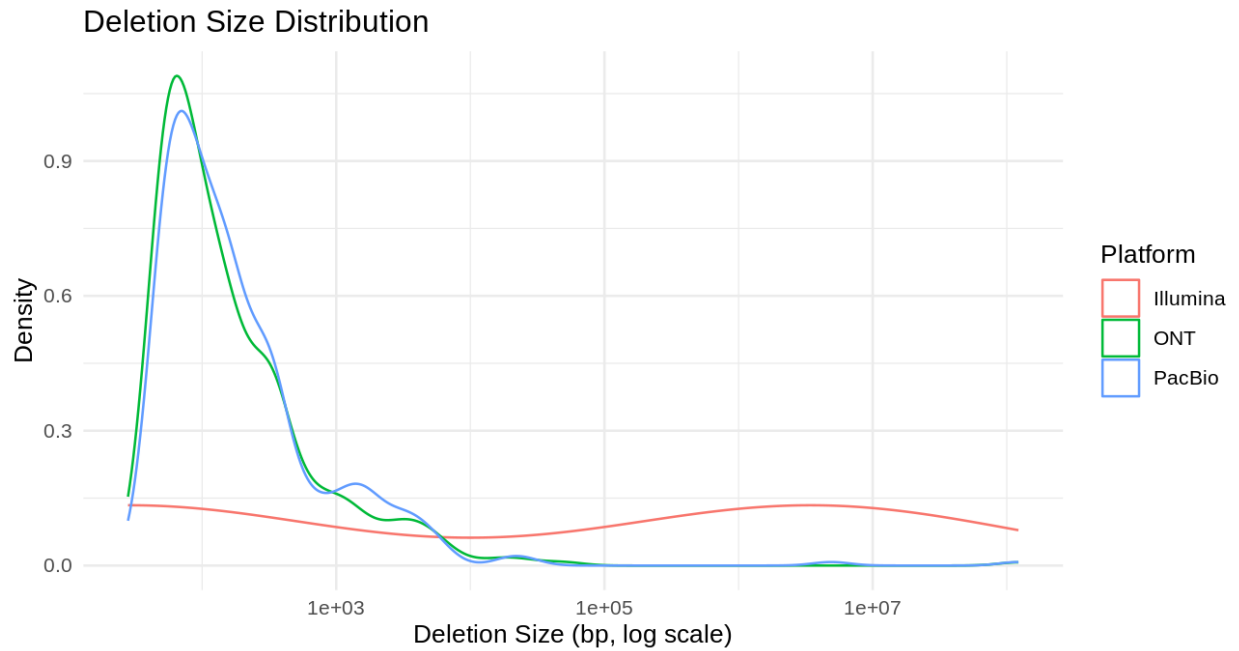
bcftools index $OUTDIR/pacbio_chr20.sniffles.pass.DEL_INS.vcf.gz

```

2. What is the minimum and maximum deletion size for Illumina, Nanopore and PacBio? Plot the density distributions of deletions for Illumina, Nanopore and PacBio in a single figure.

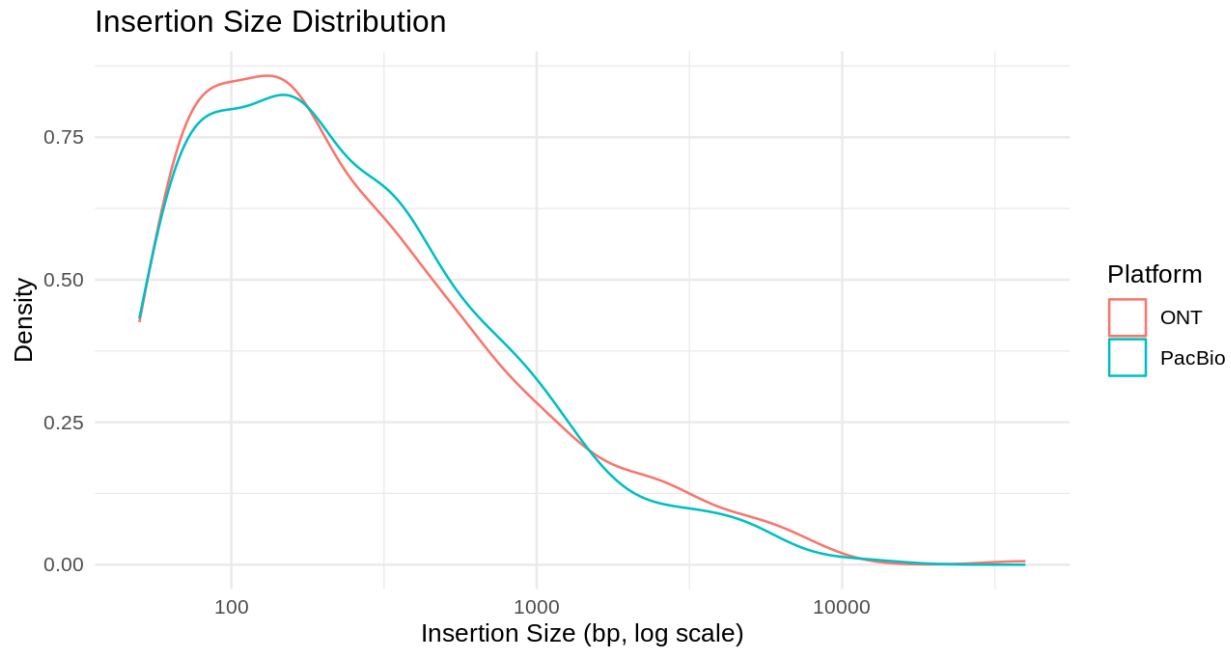
File	Min DEL	Max DEL
illumina	28 bp	3,549,528 bp
pacbio	45 bp	122,091,830 bp

File	Min DEL	Max DEL
ont	45 bp	122,091,833 bp



3. What is the minimum and maximum insertion size for Illumina, Nanopore and PacBio? Plot a the density distributions of deletions for Illumina, Nanopore and PacBio in a single figure.

File	Min INS	Max INS
illumina	None	None
pacbio	50 bp	12,225 bp
ont	50 bp	39,818 bp



Task 3: Discussion of detection methods

1. Why can we use general variant calling software that relies on alignment statistics (eg: bcftools, GATK...) to detect small Indels, but require specialized algorithms to detect large SVs?

- Bcftools and GATK detect variants by looking at base quality, mapping quality and the number of reads contribute to a variant at a position in the alignment to the reference genome.
- Small indels are a local sequence changes. Small insertions and deletions are shorter than the read length, so they can be completely contained within a single read. When mapping to the reference genome, we can directly detect how many base pairs are deleted from the alignment.
- In contrast, large structural variants involve rearrangement of an entire genomic segments, such as large deletions, insertions, or duplications. Large SVs cannot be fully represented within a single read alignment, so they require specialized algorithms to detect them.

Task 4 - Attach your code to calculate the FN and FP to the assignment. Briefly explain how the code works and why it calculates FN and FP. (10pts)

Illumina

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
```

```

#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --job-name=illumina_benchmarking
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/illumina_benchmarking%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/illumina_benchmarking%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/trgn515

IN=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/illumina_chr20
BENCH=/project2/msalomon_1816/trgn_515/2_sv/benchmark/HG002_GRCh37_SVs_Tier1_v0.6.vcf.gz
GENOME=/project2/msalomon_1816/trgn_515/2_sv/ref_genome/GRCh37/GRCh37.genome
OUT=/project2/msalomon_1816/ranzhu/assignment_2/benchmark/illumina
mkdir -p $OUT

bcftools view -i 'INFO/SVTYPE="DEL" || INFO/SVTYPE="INS"' \
$IN/illumina_chr20.sv.pass.vcf.gz \
| bcftools query -f '%CHROM\t%POS0\t%END\n' \
| bedtools slop -g $GENOME -b 100 \
> $OUT/sample.bed

bcftools view -i 'INFO/SVTYPE="DEL" || INFO/SVTYPE="INS"' \
$BENCH \
| bcftools query -f '%CHROM\t%POS0\t%END\n' \
| bedtools slop -g $GENOME -b 100 \
> $OUT/benchmark.bed

#TP
bedtools intersect \
-a $OUT/sample.bed \
-b $OUT/benchmark.bed \
-f 0.5 -r -u \
> $OUT/TP.bed

# FP
bedtools intersect \
-a $OUT/sample.bed \
-b $OUT/benchmark.bed \
-f 0.5 -r -v \
> $OUT/FP.bed

# FN
bedtools intersect \
-a $OUT/benchmark.bed \
-b $OUT/sample.bed \
-f 0.5 -r -v \
> $OUT/FN.bed

```

pacbio

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --job-name=pacbio_benchmarking
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/pacbio_benchmarking%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/pacbio_benchmarking%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/trgn515

IN=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/pacbio_chr20
SAMPLE_VCF=$IN/pacbio_chr20.sniffles.pass.vcf.gz

BENCH=/project2/msalomon_1816/trgn_515/2_sv/benchmark/HG002_GRCh37_SVs_Tier1_v0.6.vcf.gz
GENOME=/project2/msalomon_1816/trgn_515/2_sv/ref_genome/GRCh37/GRCh37.genome

OUT=/project2/msalomon_1816/ranzhu/assignment_2/benchmark/pacbio
mkdir -p $OUT

bcftools view -i 'INFO/SVTYPE="DEL" || INFO/SVTYPE="INS"' $SAMPLE_VCF \
| bcftools query -f '%CHROM\t%POS0\t%END\n' \
| bedtools slop -g $GENOME -b 100 \
> $OUT/sample.bed

bcftools view -i 'INFO/SVTYPE="DEL" || INFO/SVTYPE="INS"' $BENCH \
| bcftools query -f '%CHROM\t%POS0\t%END\n' \
| bedtools slop -g $GENOME -b 100 \
> $OUT/benchmark.bed

bedtools intersect -a $OUT/sample.bed -b $OUT/benchmark.bed -f 0.5 -r -u > $OUT/TP.bed
bedtools intersect -a $OUT/sample.bed -b $OUT/benchmark.bed -f 0.5 -r -v > $OUT/FP.bed
bedtools intersect -a $OUT/benchmark.bed -b $OUT/sample.bed -f 0.5 -r -v > $OUT/FN.bed
```

ONT

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --job-name=ont_benchmarking
```

```

#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_2/logs/ont_benchmarking%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_2/logs/ont_benchmarking%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/trgn515

IN=/project2/msalomon_1816/ranzhu/assignment_2/03_sv/ont_chr20GRCh37
SAMPLE_VCF=$IN/ont_chr20.sniffles.pass.vcf.gz
BENCH=/project2/msalomon_1816/trgn_515/2_sv/benchmark/HG002_GRCh37_SVs_Tier1_v0.6.vcf.gz
GENOME=/project2/msalomon_1816/trgn_515/2_sv/ref_genome/GRCh37/GRCh37.genome
OUT=/project2/msalomon_1816/ranzhu/assignment_2/benchmark/ont
mkdir -p $OUT

bcftools view -i 'INFO/SVTYPE="DEL" || INFO/SVTYPE="INS"' $SAMPLE_VCF \
| bcftools query -f '%CHROM\t%POS0\t%END\n' \
| bedtools slop -g $GENOME -b 100 \
> $OUT/sample.bed

bcftools view -i 'INFO/SVTYPE="DEL" || INFO/SVTYPE="INS"' $BENCH \
| bcftools query -f '%CHROM\t%POS0\t%END\n' \
| bedtools slop -g $GENOME -b 100 \
> $OUT/benchmark.bed

bedtools intersect -a $OUT/sample.bed -b $OUT/benchmark.bed -f 0.5 -r -u > $OUT/TP.bed
bedtools intersect -a $OUT/sample.bed -b $OUT/benchmark.bed -f 0.5 -r -v > $OUT/FP.bed
bedtools intersect -a $OUT/benchmark.bed -b $OUT/sample.bed -f 0.5 -r -v > $OUT/FN.bed

```

- The bedtools intersect command compares two variant files and identifies which variants overlap.
- When the sample file -a, the -v reports variants in the sample that do not overlap with the benchmark variants. These non-overlapping variants are considered False Positives, because they are detected in the sample but not present in the benchmark dataset.
- When the benchmark file -a, the -v reports variants in the benchmark that do not overlap with the sample variants. These variants are considered False Negatives, because they are present in the benchmark but were not detected in the sample dataset.
- The -f 0.5 -r option requires at least 50% reciprocal overlap, so both variants must overlap each other by at least 50%, allowing for small breakpoint differences.

Task5 - The latest task consists of a comparison between the variant calls you obtained and the gold standard.

1. How many indels are detected for each sequencing technology and how many are in the gold standard? (5pts)

- Illumina indels detected : 19981
- Pacbio indels detected: 6743
- Ont indels detected: 53747
- Gold standard vcf: 1711

2. How many large deletions are detected and how many are in the gold standard? (5pts)

- Illumina detected 2 large deletions
- Pacbio detected 372 large deletions
- Ont detected 390 large deletions
- Gold standard detected 862 large deletions

3. How many large insertions are detected and how many are in the gold standard? (5pts)

- Illumina detected 0 large insertions
- Pacbio detected 512 large insertions
- Ont detected 501 large insertions
- Gold standard detected 950 large insertions

4. How many true positive, false positive and false negative large deletions are present for each sequencing technology? (5pts)

- Illumina: TP=0, FP=2, FN=136
- Pacbio: TP=198, FP=174 FN=601
- ONT: TP= 124, FP=266, FN=13

5. How many true positive, false positive and false negative large insertions are present for each sequencing technology? (5pts)

- Illumina: TP=0, FP=0, FN=227
- Pacbio: TP=330, FP=182 FN=554
- ONT: TP= 199, FP=302, FN=29

6. Calculate the precision, recall and F1 score for indels for each sequencing technology (5pts)

illumina

- TP=13107, FP=6874, FN=337
- Precision = $13107 / (13107 + 6874) = 13107/19981 = 65.6\%$
- Recall = $13107 / (13107 + 337) = 13107/13444 = 97.5\%$
- F1 = $2 \times (0.656 \times 0.975) / (0.656 + 0.975) = 78.6\%$

pacbio

- TP=4521, FP=2222, FN=8089
- Precision = $4521 / (4521 + 2222) = 4521/6743 = 67.1\%$
- Recall = $4521 / (4521 + 8089) = 4521/12610 = 35.9\%$
- F1 = $2 \times (0.671 \times 0.359) / (0.671 + 0.359) = 46.6\%$

ont

- $TP=10129, FP=43618, FN=2498$
- $Precision = 10129 / (10129+43618) = 10129/53747 = 18.8\%$
- $Recall = 10129 / (10129+2498) = 10129/12627 = 80.2\%$
- $F1 = 2 \times (0.188 \times 0.802) / (0.188 + 0.802) = 30.6\%$

7. Calculate the precision, recall and F1 score for large deletions for each sequencing technology (5pts)

illumina

- $Precision = 0 / (0+2) = 0\%$
- $Recall = 0 / (0+136) = 0\%$
- $F1 = 0$

pacbio

- $Precision = 198 / (198+174) = 198/372 = 53.2\%$
- $Recall = 198 / (198+601) = 198/799 = 24.8\%^*$
- $F1 = 2 \times (0.532 \times 0.248) / (0.532 + 0.248) = 33.6\%$

ont

- $Precision = 124 / (124+266) = 124/390 = 31.8\%$
- $Recall = 124 / (124+13) = 124/137 = 90.5\%$
- $F1 = 2 \times (0.318 \times 0.905) / (0.318 + 0.905) = 47.1\%$

8. Calculate the precision, recall and F1 score for large insertions for each sequencing technology (5pts)

illumina

- $Precision = 0$
- $Recall = 0 / (0+227) = 0\%$
- $F1 = 0$

pacbio

- $Precision = 330 / (330+182) = 330/512 = 64.5\%$
- $Recall = 330 / (330+554) = 330/884 = 37.3\%$
- $F1 = 2 \times (0.645 \times 0.373) / (0.645 + 0.373) = 47.4\%$

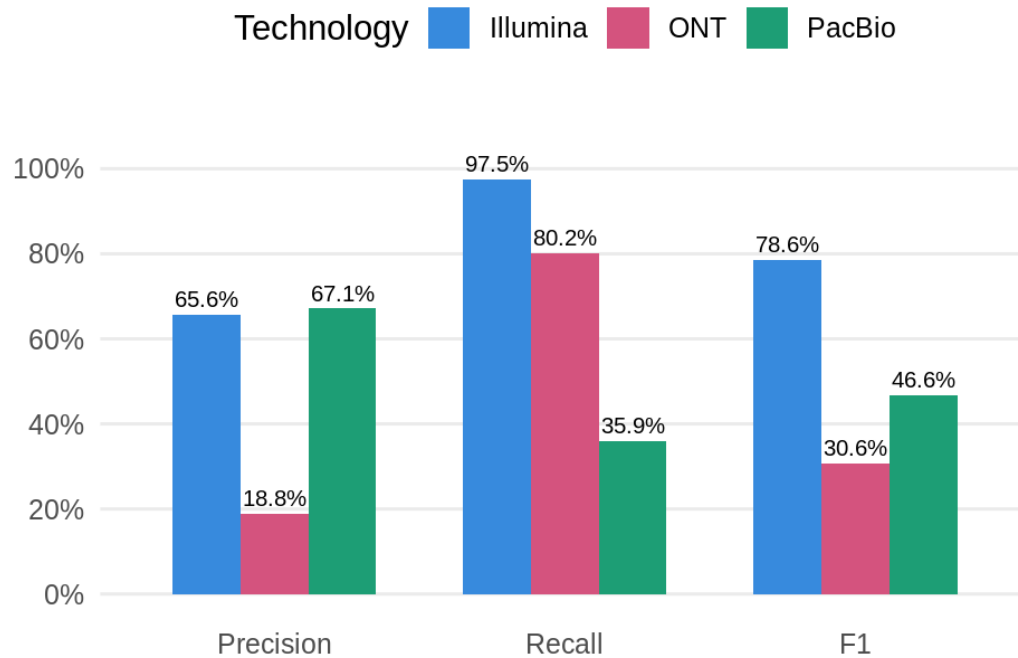
ont

- $Precision = 199 / (199+302) = 199/501 = 39.7\%$
- $Recall = 199 / (199+29) = 199/228 = 87.3\%$
- $F1 = 2 \times (0.397 \times 0.873) / (0.397 + 0.873) = 54.6\%$

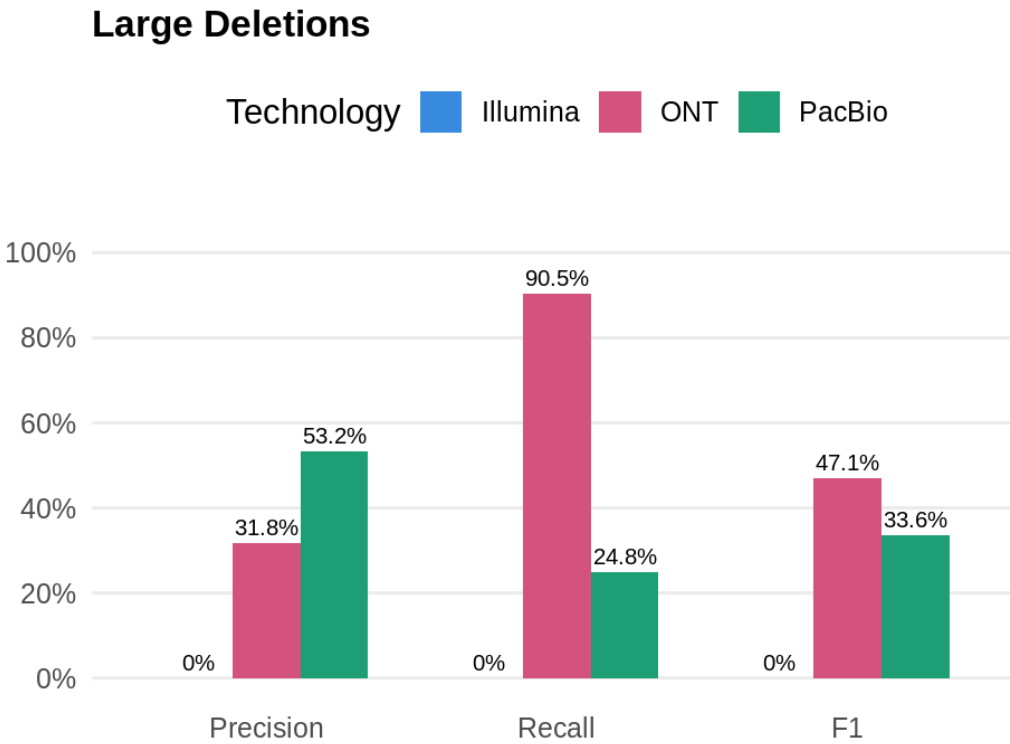
9. Make the following grouped barplots. Group bars by metric, so for each metric you can have together the three sequencing technologies (35pts).

1.Barplot for indels showing the precision, recall and F1 score

Indels



2. Barplot for large deletions showing the precision, recall and F1 score



3. Barplot for large insertions showing the precision, recall and F1 score

